

A Supercharge-Gated Correction to the Dyer/NHNE Non-Equilibrium Parameter for Self-Pressurizing N₂O Injectors:

An Exploratory Finding and Its Interaction with Discharge-Coefficient Calibration

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Abstract

The Dyer (NHNE) two-phase injector model, already implemented and validated in this project against the Waxman (2013) dataset, is known to over-predict oxidiser mass flow systematically as tank supercharge (subcooling margin above saturation pressure) decreases – mean absolute percentage error (MAPE) rises from $\approx 2\%$ at supercharge ≥ 200 psi (≈ 13.8 bar, 1.38 MPa) to $\approx 15\%$ below 100 psi (≈ 6.9 bar, 0.69 MPa), across 64 digitised two-phase points from Waxman’s injector-3 mass-flow map. A *gated* correction to Dyer’s non-equilibrium parameter κ is proposed, leaving every prediction at or above a supercharge threshold mathematically unchanged, and reducing κ below it by a power law in the ratio of the actual supercharge to the threshold. The correction is directly motivated by a limitation of κ already identified in the model’s own source literature (Vargas Niño & Razavi, 2019), who report that κ “loses its physical interpretation” at saturation, and who apply a structurally analogous supercharge-gated blend to a different two-phase model (their Omega formulation). An empirical finding not previously documented is also reported: the four already-published validation points (Niño & Razavi’s Table 4, injector 2, $\Delta P = 8\text{--}14$ bar) are not cleanly separable from the discharge-coefficient assumption used to evaluate them – with a generic $C_d = 0.65$, the gated correction degrades this subset’s MAPE from 2.75% to 4.77%; with the geometry-specific $C_d = 0.681$ measured directly by Waxman (his Fig. 15), the same correction improves it, to 1.14%. This is presented as an exploratory finding: a plausible mechanism, a working correction, one supporting empirical interaction, and an explicit list of what remains unverified.

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1 Motivation

The Dyer/NHNE model [1] blends the SPI (*single-phase incompressible*) and HEM (*homogeneous equilibrium model*) mass-flow predictions through a non-equilibrium weighting parameter,

$$\kappa = \sqrt{\frac{P_{up} - P_{down}}{P_{sat}(T_{up}) - P_{down}}}, \quad \dot{m}_{Dyer} = \frac{\kappa}{1 + \kappa} \dot{m}_{SPI} + \frac{1}{1 + \kappa} \dot{m}_{HEM}, \quad (1)$$

using the corrected weight convention of Solomon [2] and Waxman [3]. This project’s own extended validation against Waxman’s digitised injector-3 mass-flow map (Fig. 13 of [3], 64 two-phase points across nine supercharge levels) shows that Eq. (1), evaluated exactly as published, produces a mean absolute percentage error (MAPE) that depends strongly on tank *supercharge*,

$$S \equiv P_{up} - P_{sat}(T_{up}), \quad (2)$$

rather than on the injector pressure drop itself (Table 1). The error is one-directional: the model over-predicts at every point in the low-supercharge bands, never under-predicts.

Table 1: Baseline Dyer model error by tank supercharge band (Waxman injector 3, $n = 64$ two-phase points, pooled $C_d = 0.781$).

Supercharge band	n	Mean error	MAPE
< 100 psi (< 6.9 bar)	19	+15.27%	15.51%
100–200 psi (6.9–13.8 bar)	13	+5.99%	6.24%
≥ 200 psi (≥ 13.8 bar)	32	+1.91%	1.97%

This paper documents an exploratory correction targeting this specific trend, the primary-source support that motivated it, an empirical interaction with discharge-coefficient calibration discovered while testing it, and – deliberately – what is *not* established by this work.

2 Why κ Degrades at Low Supercharge: Primary-Source Support

Equation (1) is derived, and its physical interpretation discussed, in Vargas Niño & Razavi (2019) [4] – already the source of this project’s original four-point validation subset (their Table 4). Their

Eq. (4) is identical, term for term, to Eq. (1) above. Discussing its behaviour as the upstream state approaches saturation, the authors state explicitly:

“At saturated conditions the NHNE model loses its physical interpretation as the value of κ is unity and the resulting mass flux... is simply an equal averaging of the SPI and HEM models.”

This is a statement from the model’s own originating literature that κ degenerates to a supercharge-independent value ($\kappa = 1$) exactly in the regime where Table 1 shows the largest error – not an inference unique to this project.

The same paper goes further: for a *different* two-phase model (their “Omega” formulation, an equation-of-state-based approach distinct from Dyer/NHNE), they introduce an explicit supercharge threshold, η_{st} , that classifies each operating point as *high-* or *low-supercharged*, and apply, in the low-supercharge branch, a blend

$$G = \left(\frac{P_{sat}}{P_i} \right) G_{sat} + \left(1 - \frac{P_{sat}}{P_i} \right) G_{low}, \quad (3)$$

justified explicitly by “the sharp edge effect increasing the tendency for cavitation, which is more prevalent in the predictions of the saturated mass flux equation.” This is direct precedent, in the source literature this project already cites for its four-point validation subset, for a regime-gated correction to a two-phase injector model at low supercharge – applied here to a *different* model (Dyer’s κ) than the one Vargas Niño & Razavi corrected, since re-deriving their Omega formulation is a substantially larger undertaking outside this work’s scope.

Waxman (2013) [3] independently frames the same non-equilibrium behaviour in terms of a mismatch between the finite rate of interphase heat and mass transfer and the fluid’s residence time in the orifice – consistent with, though less specific than, the saturation-degeneracy argument above.

3 Proposed Correction

A gated modification to κ is introduced:

$$\kappa'' = \begin{cases} \kappa, & S \geq S_{ref} \\ \kappa \cdot \left(\frac{S}{S_{ref}} \right)^\beta, & S < S_{ref} \end{cases} \quad (4)$$

with two free parameters: the exponent $\beta \geq 0$ and the threshold supercharge S_{ref} . A power law is used, rather than an exponential or another two-parameter attenuating form, because it is the simplest such form satisfying $\kappa''(S=0) = 0$ together with independent control of curvature via β . The limit $S \rightarrow 0$ corresponds to $P_{up} \rightarrow P_{sat}(T_{up})$ – the tank sitting exactly on the saturation curve, with no subcooling margin at all – where the non-equilibrium picture underlying κ (a finite bubble-growth delay relative to residence time) is least applicable and the flow is expected to approach full thermodynamic equilibrium (the HEM limit, $\kappa \rightarrow 0$) fastest; the power-law form drives κ'' to zero exactly at this limit, consistent with that expectation.

Two properties follow directly from Eq. (4) and are worth stating precisely, since they bound what this correction can and cannot do to previously published results.

3.1 Exact recovery at $\beta = 0$ and above the threshold

For any $S \geq S_{ref}$, $\kappa'' = \kappa$ *exactly* – not approximately – regardless of β . Likewise, $\beta = 0$ gives $\kappa'' = \kappa$ everywhere. Both are immediate from Eq. (4): the correction is inert unless both conditions ($S < S_{ref}$ and $\beta > 0$) hold simultaneously.

3.2 Continuity at the threshold

Taking one-sided limits as $S \rightarrow S_{ref}$:

$$\lim_{S \rightarrow S_{ref}^-} \kappa'' = \kappa \cdot 1^\beta = \kappa, \quad \lim_{S \rightarrow S_{ref}^+} \kappa'' = \kappa, \quad (5)$$

so $\kappa''(S)$, and therefore $\dot{m}_{Dyer}(S)$ computed from it, is continuous (C^0) across the gate for any $\beta \geq 0$. This rules out a discontinuous jump in predicted mass flow as an operating point crosses S_{ref} – relevant given that this project’s coupled tank-line-injector solver (`full_system.py`) is already known to be sensitive to model discontinuities at a different threshold (the Dyer-to-HEM switch at the feed-line flashing onset; see the project’s own documentation on this point).¹

3.3 Monotonicity below the threshold

For $S < S_{ref}$ and fixed $\beta, S_{ref} > 0$, $\kappa''(S)$ is a strictly increasing power-law function of S (since $d\kappa''/dS = \kappa\beta S_{ref}^{-\beta} S^{\beta-1} > 0$ for $S > 0$), and because the Dyer blend in Eq. (1) is itself monotonic in κ for fixed SPI/HEM endpoints, $\dot{m}_{Dyer}(\kappa'')$ is monotonic in S within the gated region. No spurious local extrema are introduced by the correction inside $(0, S_{ref})$.

4 Results Against the Digitised Waxman Dataset

4.1 Grid search over the two free parameters

C_d is calibrated once, pooled over the single-phase window of all nine Waxman injector-3 curves ($C_d = 0.781$, unchanged from this project’s existing validation), independently of β and S_{ref} . A grid search over $\beta \in [0, 3.0]$ and $S_{ref} \in [50, 300]$ psi (3.4–20.7 bar, 0.34–2.07 MPa) against the 64 two-phase points yields two representative operating points (Table 2).

Table 2: Two representative points from the (β, S_{ref}) grid search. “Protects Part A” means S_{ref} sits below Part A’s own supercharge (≈ 95 psi ≈ 6.55 bar, 655 kPa), so $\kappa'' = \kappa$ exactly for those four points, by construction.

Variant	S_{ref} [psi]	β	Global MAPE	Part A MAPE	Protects Part A?
Baseline (no correction)	–	0	6.86%	2.75%	–
Conservative	80	2.5	4.63%	2.75%	Yes
Unrestricted	300	1.0	1.86%	4.77%	No

¹Continuity here is C^0 only, not C^1 : the derivative $d\kappa''/dS$ equals $\kappa\beta/S_{ref}$ immediately below S_{ref} and 0 immediately above it, so a slope discontinuity (a kink) exists at the threshold even though the value itself does not jump. This is judged an acceptable trade for the simplicity of the piecewise form in Eq. (4); a Jacobian-based solver iterating close to S_{ref} could in principle be more sensitive there than elsewhere to step size, though this has not been observed with `full_system.py`’s damped fixed-point iteration.

The unrestricted variant reduces global MAPE by a factor of $3.7\times$ relative to the uncorrected model, but at the cost of a 2.02 percentage point increase in the four-point Part A subset’s own MAPE.

4.2 An important structural fact about Part A’s own supercharge

Part A’s four points (Niño & Razavi’s Table 4, injector 2, $\Delta P = 8\text{--}14$ bar) have a tank supercharge of only ≈ 95 psi (≈ 6.55 bar, 655 kPa) – inside the low-supercharge regime this correction targets, not safely above it. Any $S_{ref} \geq 95$ psi therefore *necessarily* perturbs Part A; the “protects Part A” column of Table 2 is only available for S_{ref} values that are themselves too low to reach the 100–200 psi band, where the baseline error is also elevated (Table 1). No choice of S_{ref} corrects the 100–200 psi band while leaving Part A completely untouched: this is a structural property of the dataset, not a shortcoming of the grid search.

5 Interaction With Discharge-Coefficient Calibration

Part A’s four points were originally evaluated (both in this project and in [4]) with $C_d = 0.65$, a generic value. Waxman’s own Fig. 15 [3] reports a directly *measured* $C_d = 0.681$ for this specific injector-2 geometry (digitised as part of this project’s existing validation pipeline). Table 3 isolates the effect of this substitution, with and without the gated κ correction (using the “unrestricted” parameters of Table 2, $\beta = 1.0$, $S_{ref} = 300$ psi).

Table 3: Part A (4 points) MAPE under all four combinations of assumed C_d and the κ correction.

	No κ'' correction	With κ'' correction
$C_d = 0.65$ (generic)	2.75%	4.77%
$C_d = 0.681$ (Waxman Fig. 15, measured)	4.47%	1.14%

With the measured C_d , the gated correction does not degrade Part A – it improves it substantially, below even the originally published baseline. Neither factor in isolation explains the improvement: the measured C_d alone worsens the uncorrected baseline (2.75% $\rightarrow$ 4.47%), and the correction alone (with the generic C_d) also worsens it (2.75% $\rightarrow$ 4.77%). Only the combination improves it. This should be read as an interaction, not as one effect “explaining away” the other.

5.1 Independent support for treating C_d and supercharge regime as non-separable

Vargas Niño & Razavi [4], discussing discharge coefficients directly (their Section on “Discussion of Discharge Coefficients”), report that the appropriate C_d differs by supercharge regime – $C_d \approx 0.73$ for high supercharge (from a cavitation-number correlation) versus $C_d \approx 0.85$ suggested for low supercharge – and note that their own regime-corrected Omega model (Eq. (3)) already absorbs part of this difference, making a third, dedicated C_d unnecessary. This is independent, primary-source support for the qualitative finding of Table 3: C_d and the supercharge regime are not fully separable effects on measured mass flow.

6 A First-Order Sensitivity Argument (Not a Derivation)

The interaction reported in Section 5 can be given an approximate, order-of-magnitude account by linearising the Dyer prediction in its two calibration inputs, C_d and κ'' . Writing the Dyer blend

explicitly,

$$\dot{m}_{Dyer} = \frac{C_d A \sqrt{2\Delta P}}{\kappa'' + 1} (\kappa'' \sqrt{\rho_l} + \sqrt{\rho_{HEM}}), \quad (6)$$

and differentiating exactly (no simplifying assumption on the density ratio is required) gives

$$\frac{\partial \ln \dot{m}_{Dyer}}{\partial \kappa''} = \frac{\sqrt{\rho_l} - \sqrt{\rho_{HEM}}}{(\kappa'' + 1) (\kappa'' \sqrt{\rho_l} + \sqrt{\rho_{HEM}})}, \quad (7)$$

which is positive whenever $\rho_l > \rho_{HEM}$ (always true once vapour is present, since ρ_{HEM} mixes in the much lighter vapour phase), confirming algebraically that $\dot{m}_{Dyer}$ increases with κ'' : physically, a larger κ shifts the Dyer blend toward the SPI limit, which predicts more flow than HEM. Equivalently, *decreasing* κ'' – the direction the gated correction in Eq. (4) imposes below S_{ref} – decreases $\dot{m}_{Dyer}$, consistent with the correction’s purpose of reducing Dyer’s systematic over-prediction at low supercharge. Since $\dot{m}_{Dyer}$ is also proportional to C_d at fixed κ'' ($\partial \ln \dot{m}_{Dyer} / \partial \ln C_d = 1$ exactly), a first-order Taylor expansion of $\ln \dot{m}_{Dyer}$ in both inputs around a nominal operating point $(C_{d,0}, \kappa_0'')$ gives

$$\frac{d\dot{m}_{Dyer}}{\dot{m}_{Dyer}} = \frac{dC_d}{C_d} + \left. \frac{\partial \ln \dot{m}_{Dyer}}{\partial \kappa''} \right|_0 d\kappa'' \equiv \frac{dC_d}{C_d} + C_\phi \frac{d\kappa''}{\kappa_0''}, \quad (8)$$

where $C_\phi \equiv \kappa_0'' \left. \frac{\partial \ln \dot{m}_{Dyer}}{\partial \kappa''} \right|_0 > 0$ collects the density terms in Eq. (7) evaluated at the nominal point.

The predicted mass flow $\dot{m}_{Dyer}$ is not itself the quantity reported as a percentage error; the relative error against experiment is $E \equiv (\dot{m}_{Dyer} - \dot{m}_{exp}) / \dot{m}_{exp}$. Near a calibration point where the model already tracks the data closely ($\dot{m}_{Dyer} \approx \dot{m}_{exp}$, i.e. $|E_0| \ll 1$), a small change $d\dot{m}_{Dyer}$ shifts E by $dE = d\dot{m}_{Dyer} / \dot{m}_{exp} \approx d\dot{m}_{Dyer} / \dot{m}_{Dyer}$, since $\dot{m}_{Dyer} / \dot{m}_{exp} \rightarrow 1$ in that limit. Substituting Eq. (8), the relative error itself shifts approximately as

$$\Delta E \approx \frac{\Delta C_d}{C_{d,0}} + C_\phi \frac{\Delta \kappa''}{\kappa_0''}. \quad (9)$$

Equation (9) makes explicit why the two calibration choices are not separable in Table 3: an error sourced from C_d (first term) and an error sourced from κ'' (second term) enter the total error additively at first order, so a poorly chosen C_d and a κ correction calibrated against data evaluated with that same poor C_d will, in general, partially compensate for one another – consistent with what Table 3 shows numerically (correcting only one of the two inputs makes the error worse, correcting both together makes it much better).

This argument is presented as a linearised sensitivity statement, not a closed-form proof of the observed interaction. It is not claimed, and Table 3 does not require, that the true residual error follows the specific power-law form of Eq. (4) exactly: Eq. (9) is a first-order local statement about how small changes in C_d and κ'' move the prediction, adequate for explaining the direction and approximate scale of the interaction observed empirically, not a derivation of the specific β or S_{ref} values that fit best. Deriving those from first principles (e.g. from a bubble-growth or nucleation-delay model) is identified as future work in Section 8.

7 What This Work Does *Not* Establish

Stated explicitly, and in the same spirit as this project’s existing validation reports:

- β and S_{ref} are empirically fitted to this project’s own digitised dataset, not derived from a primary source the way κ itself, or this project’s existing Henry-Fauske ceiling, are. They should not be treated as universal constants.
- The favourable C_d -corrected result (MAPE 1.14%, Table 3) rests on only four points (Part A). This is not enough data to claim general validity for either the correction or the interaction with C_d . A genuine robustness check would apply the same β and S_{ref} (fitted on injector 3) to an independent, larger set of low-supercharge points from a *different* injector geometry – data that does not currently exist in this project’s digitised set.
- The mechanism proposed by Vargas Niño & Razavi (sharp orifice-edge geometry increasing cavitation tendency) is **geometry-specific**; whether it transfers cleanly to a correction of κ specifically, as opposed to their own Omega formulation, has not been checked here against CFD or a first-principles nucleation model – only against aggregate mass-flow data.
- No claim is made about two-phase choking or the effective speed of sound in the metastable mixture. A possible connection to choked-flow behaviour has not been tested against this project’s choking diagnostics (Henry-Fauske, HEM critical flow) and is therefore outside the scope of this work.
- This correction is not part of the tool’s production path. `full_system.py` continues to call the original, unmodified `dyer_mass_flow()`; the gated variant (`dyer_mass_flow_corrected()`) exists as an additional, standalone function, following the same diagnostic-first discipline already used for the Henry-Fauske ceiling in this project.

8 Future Work

1. **Independent geometry.** Digitise or acquire a second injector’s low-supercharge two-phase data to test whether β and S_{ref} fitted on injector 3 generalise, or whether they are specific to this geometry.
2. **First-principles threshold.** Investigate whether S_{ref} can be related to a physically motivated time-scale ratio (residence time in the orifice versus a bubble-growth or nucleation-delay time scale), rather than fitted directly – analogous to how this project’s existing Henry-Fauske ceiling is derived from first principles rather than fitted.
3. **Decoupled C_d calibration.** Given Section 5, recalibrate C_d per injector geometry before any future (β, S_{ref}) grid search, rather than pooling across geometries, to avoid conflating the two effects.
4. **Promotion criterion.** As with the Henry-Fauske ceiling, define in advance the evidence threshold (e.g. number of independent low-supercharge validation points, or a first-principles derivation of S_{ref}) required before considering this correction for the production path rather than as a diagnostic.

9 Reproducibility

All figures in this report are produced by `validation/explore_supercharge_correction.py`, which reuses the existing data-loading and C_d -calibration functions of `validation/waxman.2013_validation.py` without duplicating logic. The gated correction itself is implemented as `dyer_mass_flow_corrected()`

in `src/model/injector_two_phase.py`, alongside (not replacing) the existing `dyer_mass_flow()`. See `docs/future_work.md`, Priority 1, for how this exploration is tracked relative to the project's existing Henry-Fauske diagnostic.

References

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